PROJECT CASE STUDY

Campus Exchange

AI-Powered Campus Marketplace

Category	Status	Year
Full-Stack AI	Production system	2026

A verified campus marketplace combining recommendation search, pricing intelligence, expense sharing, and production-ready APIs for safer student transactions.

15%	Full-stack	Verified
pricing accuracy gain	AI marketplace	user workflow

Context and problem

Students need a trusted place to buy and sell items, split expenses, and avoid scams without depending on fragmented social-media groups.

Strategy and implementation

- Designed a Next.js frontend backed by FastAPI services.
- Used PostgreSQL with pgvector for semantic and personalized recommendations.
- Added Redis for fast access patterns and production-oriented caching.
- Structured the product around verified users, price guidance, and practical campus workflows.

My role

Founder and lead developer - product architecture, AI features, backend services, data layer, and deployment decisions.

Technology stack

Next.js | FastAPI | PostgreSQL | pgvector | Redis | Python | JavaScript

Results and evidence

- Reported a 15% gain in pricing accuracy.
- Delivered a full-stack production system focused on verified campus users.
- Turned recommendation and pricing models into user-facing marketplace features.

Key takeaways

- AI features matter most when they reduce a real user risk.
- Vector search becomes valuable when it is integrated into a complete product journey.

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